

Practice Quiz: Systems (Mathematical Frameworks)

Name: __ Date: __

Part A: Multiple Choice

1. A probability distribution must satisfy: A) All values are between -1 and 1 B) All probabilities are between 0 and 1, and they sum/integrate to 1 C) At least one value equals 1 D) Values can be negative
2. Bayes' theorem states $P(s | o) =$: A) $P(o | s) + P(s) / P(o)$ B) $P(o | s) \times P(s) / P(o)$ C) $P(s) / P(o | s)$ D) $P(o) \times P(s)$
3. In a generative model, $P(o | s)$ represents: A) The probability of a hidden state given an observation B) How likely observation o is given hidden state s (the likelihood) C) The prior probability of the observation D) The entropy of the system
4. A Hidden Markov Model adds temporal dynamics through: A) $P(o | s)$ only B) A transition distribution $P(s_t | s_{t-1})$ that specifies how states evolve over time C) Removing all uncertainty D) Doubling the number of observations
5. The prior $P(s)$ in Bayes' theorem represents: A) What we believe after seeing the data B) What we believed before seeing any data C) The sensory data itself D) The likelihood of the model
6. In the disease-testing example, the "base rate fallacy" occurs when people: A) Correctly apply Bayes' theorem B) Ignore the low prior probability of disease and overestimate $P(\text{disease} | \text{positive test})$ C) Never take medical tests D) Require too many tests
7. A generative model $P(o, s) = P(o | s) \times P(s)$ is called "generative" because: A) It can only classify observations B) It specifies the process by which hidden states generate observations C) It generates energy D) It was invented by a geneticist

Part B: Short Answer

1. A disease affects 0.1% of the population. A test has $P(+ \mid \text{disease}) = 0.99$ and $P(+ \mid \text{no disease}) = 0.05$. You test positive. Calculate $P(\text{disease} \mid +)$ using Bayes' theorem. Explain why the result might be surprising.
2. Construct a simple generative model for a student and their grade: hidden state $s \in \{\text{studied, didn't study}\}$ and observation $o \in \{A, C, F\}$. Define $P(s)$ and $P(o \mid s)$. Use Bayes to infer: if a student gets an A, what's the probability they studied?
3. Explain intuitively why a generative model $P(o, s) = P(o \mid s) \times P(s)$ is the mathematical formalization of the "brain's model of the world" described in the Cognitive Science unit. What are s , o , and $P(o \mid s)$ in terms of the brain?